

HAT-P-18b

R-2524

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_170616 · created 2026-08-25T01:29:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457920.69574 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.053
Transit depth (Rp/Rs)^2	3.31 +/- 1.92 [%]
Orbital Inclination (inc)	89.88 +/- 1.87
Airmass coefficient 1 (a1)	8.06 +/- 3.79
Airmass coefficient 2 (a2)	-2.1 +/- 0.54
Scatter in the residuals of the lightcurve fit is	37.4 %
Best Comparison Star	#1 - [483, 166]
Optimal Aperture	2.53
Optimal Annulus	3.0
Transit Duration (day)	0.086 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.053 vs 0.1356 (deviation 0.65 σ)
Duration fit vs published	0.086 d vs 0.1129 d (deviation 1.11 σ)
Mid-time O-C	-0.4 min
Depth SNR	2.5
Residual scatter	37.4 %

Light curve

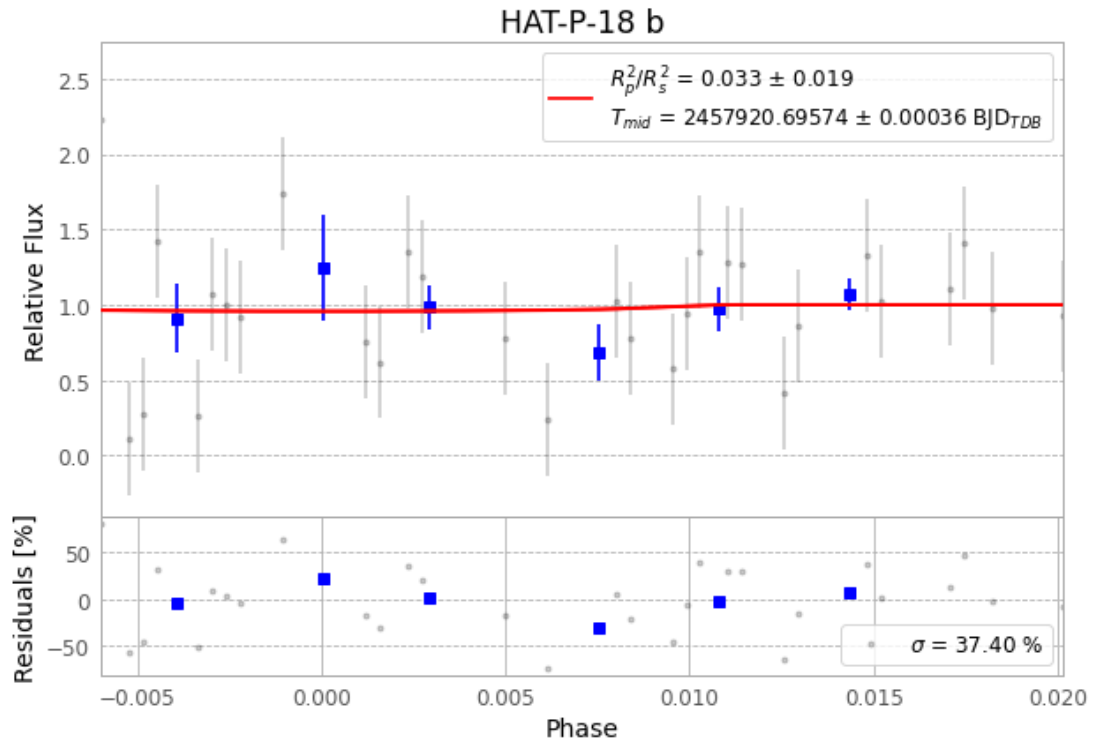


Figure 1 — FinalLightCurve_HAT-P-18 b_2017-06-15.png (SHA-256 27077cbc8bb2a295...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [120, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fc85c57655d542d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), HATP-18170616034211.FITS → HATP-18170616071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-170616.log (e50bf2f06e2c...), FinalLightCurve_HAT-P-18 b_2017-06-15.png (27077cbc8bb2...), FinalLightCurve_HAT-P-18 b_2017-06-15.csv (628bad82995e...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:29Z.